An Object Fine-Grained Change Detection Method Based on Frequency Decoupling Interaction for High-Resolution Remote Sensing Images

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***Abstract*— Change detection is a prominent research direction in the field of remote sensing image processing. However, most current change detection methods focus solely on detecting changes without being able to differentiate the types of changes, such as "appear" or "disappear" of objects. Accurate detection of change types is of great significance in guiding decision-making processes. To address this issue, this article introduces the object fine-grained change detection (OFCD) task and proposes a method based on frequency decoupling interaction (FDINet). Specifically, in order to enhance the model's ability to detect change types and improve its robustness to temporal information, a temporal exchange framework is designed. Additionally, to better capture spatial–temporal correlation in bi-temporal features, a wavelet interaction module (WIM) is proposed. This module utilizes wavelet transform for frequency decoupling, separating features into different components based on their frequency magnitudes. Then the module applies different interaction methods according to the characteristics of these frequency components. Finally, to aggregate complementary information from different-scale feature maps and enhance the representational capabilities of the extracted features, a feature aggregation and upsampling module (FAUM) is adopted. A series of experiments show the superiority of FDINet over most state-of-the-art methods, achieving good results on three different datasets.**

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***Index Terms*— Feature aggregation, frequency decoupling interaction, object fine-grained change detection (OFCD), temporal exchange.**

I. Introduction

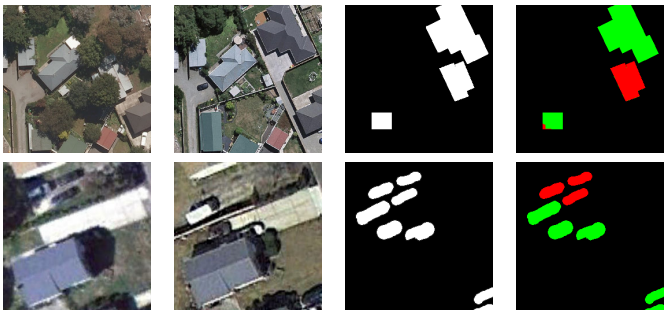
CHANGE detection, as a vital remote sensing image processing task, has witnessed remarkable advancements. Change detection primarily involves the identification and analysis of changes within the same geographical area across distinct temporal intervals [1]. It plays a crucial role in providing technical guidance for decision-making in various domains such as environmental monitor [2], [3] resource management [4], [5], and natural disaster assessment [6], [7]. Hence, the automated and precise detection of change regions can yield improved operational efficiency, labor-saving, and resource conservation.

In the early stages of research, the primary emphasis was on utilizing statistical methods, such as change vector analysis (CVA) [8] and Markov random field (MRF) models [9], to classify individual pixels and their neighboring counterparts. With the continuous improvement in the resolution of remote sensing images, coupled with the increasing complexity of geospatial information, conventional machine learning techniques have encountered difficulties in extracting image features and detecting changes effectively.

In recent years, the development of deep learning has led to the design and utilization of a growing array of deep learning models specifically tailored for change detection tasks. Daudt et al. [10] introduces a Siamese network, which enhances the model's ability to detect changes by sharing encoder weights. Based on Siamese networks, various mechanisms related to change detection tasks have been adopted to enhance the accuracy of the model. These mechanisms include deep supervision mechanisms that impose multilevel constraints on model performance [11], [12], [13], [14], multiscale mechanisms that aggregate features at different scales [11], [15], [16], [17], and attention mechanisms that expand the model's receptive field in a variable manner [18], [19], [20], [21], [22]. What's more, the rise of transformers has garnered substantial attention in the field of remote sensing imagery due to their heightened capacity for capturing long-range dependencies. Many methods have adopted transformers to extract global features from images, such as [23], [24], [25], and [26]. In addition, there are several methods that achieve change detection from unique perspectives. For example, FMCD [27]

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(a) (b) (c) (d)

Fig. 1. Illustration of the OFCD. (a) First temporal (prechange) image. (b) Second temporal (pro-change) image. (c) Traditional change detection label, where white pixels indicate the change area. (d) OFCD label, where red pixels indicate the "disappear" change area and green pixels indicate the "appear" change area.

incorporates domain adaptation to reduce the style differences between two temporal images. P2V-CD [28] treats change detection as a video understanding task and extracts temporal features from the images. CADM [29] utilizes a denoising diffusion probabilistic model (DDPM) to guide the difference map of the bi-temporal images, thereby obtaining the change detection results. Wang et al. [30] comprehensively considers different categories of changes and enhances the accuracy of street scene change detection by integrating the features of different categories of changes through a multibranch module.

While the aforementioned methods have achieved notable results in the field of change detection, the tasks performed so far have been primarily focused on detecting changes without further categorizing the types of changes. As shown in Fig. 1, most change detection methods primarily focus on detecting the changed regions. While this provides some technical support for decision-making, it fails to differentiate the types of changes, such as "appear" or "disappear." When individuals are primarily interested in the changes of particular objects, the distinction between categories of such changes is crucial in providing informative guidance. For instance, the construction or destruction of buildings, the increase or decrease in cultivated land area, and the changing trends of riverbeds, among others. We define the task of further categorizing changes pertaining to specific objectives as the object fine-grained change detection (OFCD) task.

Modeling temporal relationships contributes to distinguishing changes, both their presence and types. This is crucial for OFCD tasks. Nevertheless, most current methods encounter challenges in effectively extracting temporal contextual information. For this problem, we have observed a phenomenon: the texture information within the changing regions does not discriminate between different change categories (i.e., the boundary and the line features within the changing regions remain consistent whether the change category is "appear" or "disappear"). These textural features correspond to the high-frequency components in the image's frequency domain, whereas other features that encompass temporal correlations correspond to the low-frequency components [31]. In order to enhance the extraction of temporal features, the consideration of frequency-decoupled methods, wherein features are separated into different frequency components and processed individually, has been explored.

Therefore, a frequency decoupling interaction-based method (FDINet) is employed to address the aforementioned OFCD task. First, for the OFCD task, a temporal exchange framework is proposed, which leverages the interrelation between two types of changes to devise a change dice loss. This loss function is formulated to constrain the model's capacity to discriminate between different change categories and enhance the model's robustness to change types. Furthermore, in order to better capture the spatial–temporal correlations of input features, a wavelet interaction module (WIM) is proposed. It employs wavelet transform [32] to decouple the features into different frequency subbands and performs interactions separately based on different frequencies. Subsequently, the feature aggregation and upsampling module (FAUM) is designed to fuse and gradually upsample the different-scale features. The resulting features are then fed into a classifier to obtain the final change detection results.

The main contributions are as follows.

1) Aiming at achieving OFCD tasks, a temporal exchange framework is proposed. This framework employs a frequency-based decoupling of input features to extract temporal correlation information related to bi-temporal features, which is of paramount importance for OFCD tasks. In addition, a change dice loss is implemented to better constrain the model in distinguishing change categories and enhance the model's robustness to change types. To the best of our knowledge, this is the first attempt in the field of change detection in remote sensing to directly address the OFCD task in a supervised manner.
2) A WIM is proposed, which decouples the bi-temporal features into different frequency subbands. Based on the extent of temporal contextual information they encapsulate, distinct frequency subbands employ diverse interaction mechanisms. This approach enhances the extraction of features conducive to OFCD tasks.
3) A FAUM is designed, which aggregates the semantic information from deep-level features and the texture information from shallow-level features, thereby extracting highly discriminative features and improving detection accuracy.

The remaining sections of this article are structured as follows. Section II provides a concise overview of the relevant prior research. In Section III, we present the comprehensive details of FDINet. Section IV showcases the experimental results and analysis. Finally, Section V concludes the article.

II. Related Works

A. Binary Change Detection Method

In recent years, deep learning-based methods have gained increasing popularity in the field of change detection. Generally, change detection primarily focuses on detecting changes between bi-temporal images. In this regard, the interaction between bi-temporal features plays a crucial role. Based on the interaction approach, current change detection methods can be broadly categorized into four types: difference-based meth-

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ods, concatenation-based methods, hybrid-based methods, and other methods.

The differential-based methods primarily involve computing the absolute difference between the feature of two temporal images. This parameter-free approach effectively highlights areas of change, making it a predominant technique in change detection. FC-Siam-diff [10] is a typical difference-based method that performs differencing on each layer of the encoder's feature maps and continuously upsamples them to obtain the final result. Lv et al. [20], Chen et al. [23], Wang et al. [33], and others also follow similar principles In [12], a deep supervision strategy is employed to constrain the internal feature difference map through label constraints. What's more, Wen et al. [29] utilize the difference map of bi-temporal features as the conditional guidance for conditional diffusion models. They iteratively denoise and diffuse the guidance difference map to obtain the final change detection result.

The concatenation-based method primarily entails the concatenation of bi-temporal features along the channel dimension. In FC-EF [10], bi-temporal images are initially concatenated along the channel dimension and then fed into the network for training. LSNet-dense [34] follows a similar principle as well. SNUnet [18] and DSIFN [11] perform feature map concatenation within the model. Additionally, TinyCD [16] and FMCD [27] achieve interaction by cross-mixing bi-temporal features along the channel dimension.

The hybrid-based methods combine feature difference and feature concatenation. For example, DMINet [35] employs separate branches for the feature difference and the feature concatenation, and ultimately merges the outputs of both branches to obtain the detection results. TFI-GR [36] leverages the combination of two approaches to extract temporal information from bi-temporal features.

In addition, there are other methods such as P2V-CD [28], which treats change detection as a video understanding task, and AFCF3D-Net [37], which considers bi-temporal images as two frames and extracts temporal correlations using 3-D convolutions.

However, the difference-based methods result in the loss of temporal correlation information, as the model can only determine the areas of change through the difference maps obtained, without the ability to identify the category of the change. The concatenation-based methods can exploit the temporal information of bi-temporal features. However, due to the fact that the majority of concatenation-based methods directly operate on the channel dimension of feature maps, they introduce some redundant information unrelated to the temporal features, thus compromising the extraction capability to some extent. Therefore, to better capture the spatial–temporal correlations between bi-temporal images, this article proposes a WIM that employs different interaction methods for different frequency components of the bi-temporal features.

B. Frequency-Decoupled Method

Currently, most computer vision methods primarily process images in the spatial domain [38], [39], [40]. However, as the resolution of images continues to increase, the information within the images becomes more complex, and many methods struggle to effectively capture both local and global information of the images. Liu et al. [41] argue that semantic information is crucial during the encoding process, while detailed information is essential during the decoding process. Therefore, they enhanced the low-frequency components in the encoder and the high-frequency components in the decoder, which resulted in promising results in video semantic segmentation tasks. Furthermore, the decoupling of images based on frequency levels has also been widely applied in domain adaptation tasks. For instance, in the case of [42], [43], and [44], they decompose the images into distinct frequency subbands based on varying frequencies, subsequently categorizing them into domain-invariant features and domain-specific features. Wavelet transform not only enables lossless image resizing but also allows for the decomposition of images into different frequency components. It has been extensively utilized in the field of computer vision [45], [46]. Yang et al. [47] utilized wavelet transform to decouple images and designed a wavelet attention network, which achieved promising results in the field of image classification.

Therefore, in this article, bi-temporal images are decoupled into different frequency components using wavelet transform. Diverse interaction strategies are employed based on the varying levels of temporal correlation encapsulated within different frequency components. This approach effectively captures spatial–temporal joint features conducive to OFCD tasks.

III. Methodology

In this section, the structure of FDINet is introduced first. Subsequently, we introduce the temporal exchange framework, followed by a detailed explanation of the WIM. Finally, the FAUM is elaborated upon.

A. Overview

The overview of FDINet is illustrated in Fig. 2. As depicted in the figure, FDINet consists of two main components: the backbone and the temporal exchange framework. The temporal exchange framework comprises three essential elements: the frequency interaction part, the feature aggregation part, and the classifier.

The initial processing of images primarily focuses on shallow features such as color and lines, which are common features across all images [16]. Therefore, employing a pretrained backbone can effectively enhance the model's performance and convergence speed. In this article, the first five layers of EfficientNet-b4, pretrained on ImageNet, were employed as the backbone architecture for FDINet. The bi-temporal images $\mathcal{T}_1, \mathcal{T}_2 \in \mathbb{R}^{3\times H\times W}$ are initially encoded into feature maps of different scales f_1^i, f_2^i through the shared-weight backbone. Where H and W represent the height and width of the input image, and i denotes the number of feature map scales, $i = 1, 2, 3, 4$.

Subsequently, within the temporal exchange framework, f_1^i and f_2^i are combined to create two distinct temporal orders. By decoupling the bi-temporal features in terms of

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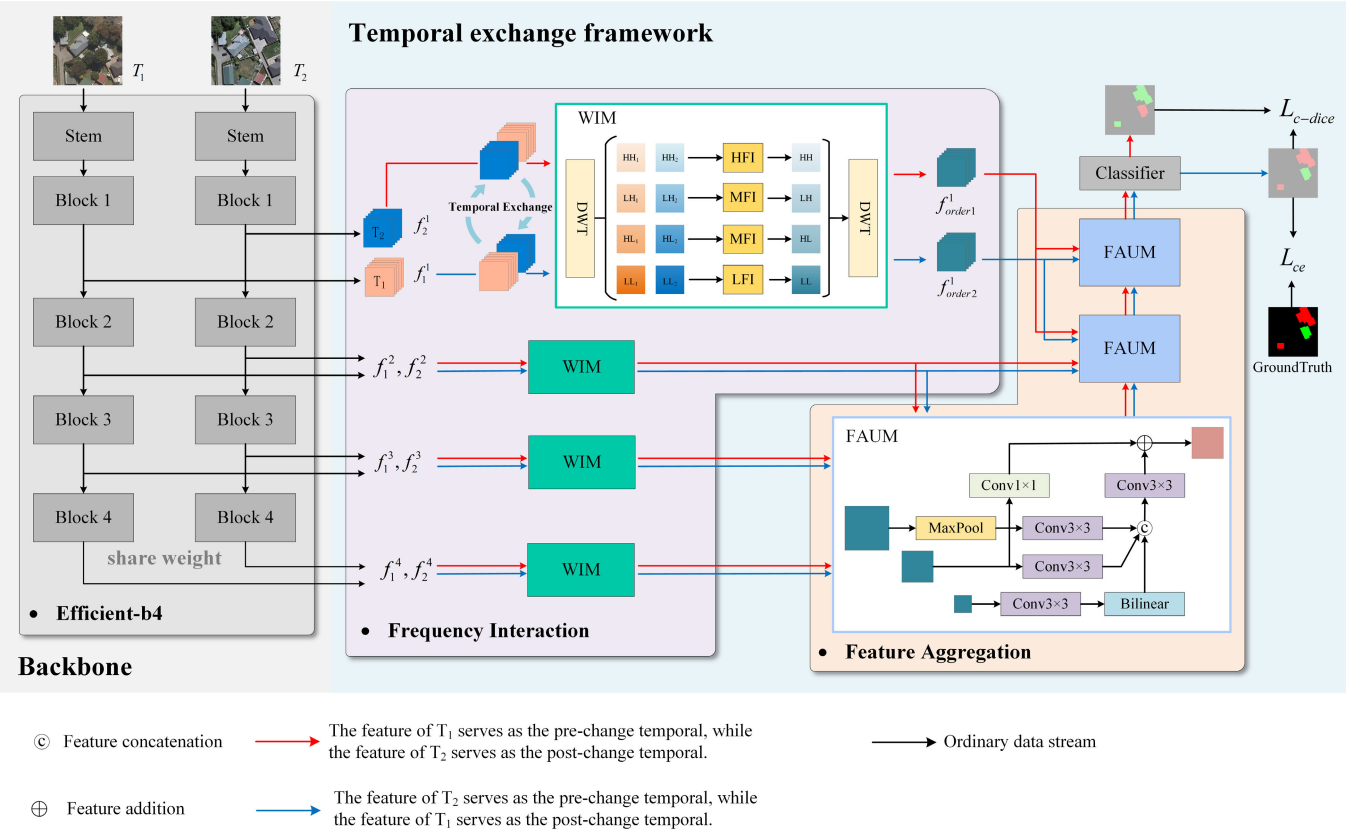


Fig. 2. Framework of FDINet. FDINet consists of two main components: a pretrained backbone and a temporal exchange framework. Within this framework, the bi-temporal features are combined using different temporal orders. After passing through the frequency-decoupling interaction of the WIM and the feature aggregation of the FAUM, the features are fed into a classifier to obtain the final results.

frequency and enabling their mutual interaction, respectively, spatial–temporal correlations are extracted. After obtaining feature maps with diverse scales and rich spatial–temporal correlations, FAUMs are established to integrate complementary information among different scale feature maps and gradually upsample them.

Ultimately, considering that OFCD tasks still fall under the category of dense segmentation tasks, the classifier is constructed by stacking three pixel-level convolutional layers. The activation function of the final convolutional layer is set as Softmax().

B. Temporal Exchange Framework

In the temporal exchange framework, as shown by the red and blue arrows in Fig. 2, the bi-temporal features f_1^i and f_2^i are initially exchanged to reverse their temporal order. Specifically, the feature representation from $\mathcal{T}_1$ is employed as the prechange temporal, while the feature representation from $\mathcal{T}_2$ is employed as the post-change temporal: $[f_1^i, f_2^i]$, and vice versa $[f_2^i, f_1^i]$.

Then, the feature maps from both temporal orders are combined and frequency-decoupled into distinct frequency components. These components with the same frequency are then interacted with each other to extract spatial–temporal joint features from the bi-temporal features (detailed in Section III-C), which is crucial for OFCD tasks. It is worth noting that the difference in the two temporal orders is specifically reflected in the interaction of the various frequency subbands after frequency decoupling. Specifically, in the OFCD tasks, the temporal information between the bi-temporal features is critical for discriminating change categories, while the spatial information is essential for enhancing detection region accuracy. However, not all features contain temporal or spatial information. For instance, texture features extracted from bi-temporal images can constrain the boundaries and semantics of the detection area, but they lack temporal relevance. The texture information in an image corresponds to its high-frequency components, while other stylistic, content-related, and similar information corresponds to its low-frequency components [41], [48]. Therefore, the temporal exchange framework employs a time-invariant interaction approach for high-frequency components that contain almost no temporal-related information. However, for low-frequency components, the temporal order is altered by changing the combination order of the bi-temporal components.

After obtaining the spatial–temporal joint features through interaction, these features of different scales are aggregated and upsampled through FAUM. Subsequently, a classifier is employed to obtain the final OFCD results. It is worth noting that each temporal order $[f_1^i, f_2^i]$ or $[f_2^i, f_1^i]$ yields a result l^1, l^2, respectively.

Considering that when the temporal order of the two features is exchanged, although the change type change, the changed regions remain the same. In other words, the change

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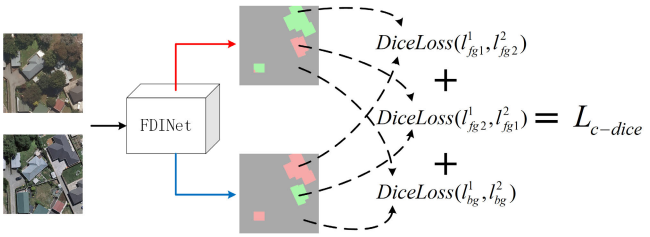


Fig. 3. Illustration of $L_{c-\text{dice}}$.

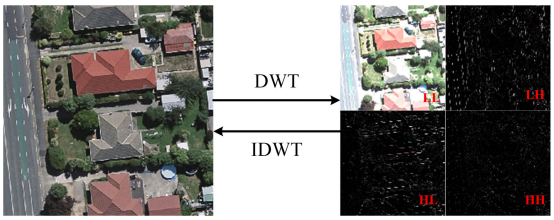


Fig. 4. Illustration of DWT and IDWT.

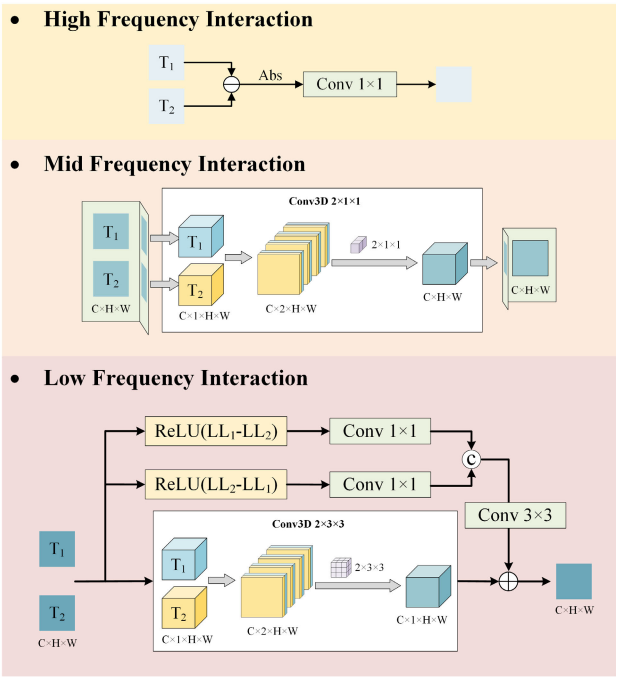


Fig. 5. Illustration of different interaction approaches. Here, T_1 and T_2 represent the corresponding frequency components of the bi-temporal features.

in temporal order only leads to a transition in change type from "appear" to "disappear" or from "disappear" to "appear." Leveraging the relationship between two types of changes, a change dice loss $L_{c-\text{dice}}$ is proposed, which exhibits the capability to enhance the model's discernment of distinct change types and to augment the model's robustness toward diverse types of changes. As illustrated in Fig. 3, in the output result of FDINet, there are three classes: background (represented as l^i_{bg}), appear (represented as l^i_{fg1}), and disappear (represented as l^i_{fg2}), where $i = 1, 2$ represents two different temporal orders. After temporal order exchanging, the background, representing the unchanged regions, remains unaffected, while the categories of the changed regions are exchanged. Therefore, in this article, Dice loss [49] is utilized to constrain the three classes accordingly. The formula for Dice loss is as follows:

$$\text{DiceLoss}(X, Y) = 1 - \frac{2|X \cap Y|}{|X + Y|}. \tag{1}$$

Specifically, l^1_{bg} and l^2_{bg} mutually constrain each other, l^1_{fg1} and l^2_{fg2} mutually constrain each other, and l^2_{fg1} and l^2_{fg1} mutually constrain each other. As is shown in formula

$$L_{c-\text{dice}}\left(l^1, l^2\right) = \text{DiceLoss}\left(l^1_{\text{bg}}, l^2_{\text{bg}}\right) + \text{DiceLoss}\left(l^1_{\text{fg1}}, l^2_{\text{fg2}}\right) + \text{DiceLoss}\left(l^2_{\text{fg1}}, l^1_{\text{fg2}}\right). \tag{2}$$

Furthermore, the cross-entropy (CE) loss L_{ce} is computed between the results l^1 generated based on the original temporal order and the dataset labels. The final loss function of FDINet is then formulated as follows:

$$L\left(l^1, l^2, \text{label}\right) = L_{\text{ce}}\left(l^1, \text{label}\right) + L_{c-\text{dice}}\left(l^1, l^2\right). \tag{3}$$

C. Wavelet Interaction Module

Discrete wavelet transform (DWT) [32] can decompose an image into multiple frequency subbands, allowing for the extraction of local features, making it widely applicable in the field of computer vision [50], [51], [52]. Therefore, in this article, the 2-D Haar DWT is employed. Specifically, the image $X \in \mathbb{R}^{C\times H\times W}$ is passed through a low-pass filter $f_{\text{low}} = [(1/\sqrt{2}), (1/\sqrt{2})]$ and a high-pass filter $f_{\text{high}} = [(1/\sqrt{2}), -(1/\sqrt{2})]$ to obtain the horizontal low-frequency X_L and high-frequency components X_H. Subsequently, the same low-pass and high-pass filters are applied to X_L and X_H in the vertical direction, resulting in four frequency subbands: $X_{LL}, X_{LH}, X_{HL}, X_{HH} \in \mathbb{R}^{C\times(H/2)\times(W/2)}$. By performing an inverse combination on these four frequency subbands, the Haar inverse DWT (IDWT) can be achieved, leading to the reconstruction of the original image X. Fig. 4 illustrates the results of applying the DWT and IDWT on a remote sensing image.

In FDINet, the multiscale bi-temporal features are first subjected to 2-D DWT, resulting in four frequency components: LL_1, LH_1, HL_1, HH_1 and LL_2, LH_2, HL_2, HH_2, respectively. Then, different frequency components are interacted using distinct approaches, as illustrated in Fig. 5. In the figure, each convolutional layer, such as $\text{Conv}_{1\times1}$, is composed of one single convolution, one batch normalization, and one activation function.

For the high-frequency component of the bi-temporal features, which primarily contains texture information and lacks temporal relevance, the simplest difference approach is employed to extract the change feature, as shown in the formula

$$HH = \text{Conv}_{1\times1}(|HH_1 - HH_2|). \tag{4}$$

For the LH and HL subbands, which contain a small amount of low-frequency information, they are referred to as mid-frequency components to distinguish them from the HH subband. Due to the mid-frequency components containing a small amount of low-frequency information, which

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also contain certain temporal correlations, a 3-D convolution is employed to extract their temporal features. Specifically, the mid-frequency subbands from the two temporal features are expanded into an additional dimension, referred to as depth $LH_1, HL_1, LH_2, HL_2 \in \mathbb{R}^{1\times C\times H\times W}$ and then concatenated along the depth dimension $LH, HL \in \mathbb{R}^{2\times C\times H\times W}$. By employing 3-D convolution, the concatenated subbands enable cross-mixing of the two temporal features in the channel dimension. Subsequently, the interaction between the bi-temporal features is achieved through a convolutional layer with a kernel size of $2 \times 1 \times 1$ and a stride of $2 \times 1 \times 1$. Note that since the LH and HL components predominantly contain high-frequency information, which corresponds to texture features, the spatial dimension's convolutional kernel size is set to 1. As shown in the formula

$$LH = \text{Conv}_{2\times1\times1}([LH_1, LH_2]) \tag{5}$$
$$HL = \text{Conv}_{2\times1\times1}([HL_1, HL_2]) \tag{6}$$

where $[\cdot, \cdot]$ denotes the operation of concatenating two inputs along the depth dimension after expanding them with an additional depth channel.

The low-frequency component contains the majority of the information in the feature maps [53]. First, to fully extract the spatial–temporal correlated information from the bi-temporal features, a similar interaction approach as in the mid-frequency is employed, utilizing a 3-D convolution. It is worth noting that in the low-frequency component, the 3-D convolutional kernel has a size of $2 \times 3 \times 3$, aiming to extract temporal features while capturing spatial structural information. Additionally, two simple branches are designed to amplify the differences between the two types of changes. Specifically, this is achieved by subtracting the dual-temporal features and taking the nonnegative portion of the difference map. While this approach cannot determine which specific type of change it is, the two branches together ensure coverage of both types of changes. Subsequently, the feature maps of the two types of changes are concatenated along the channel dimension and aggregated using a convolutional layer with a kernel size of 3, resulting in a change feature map. This change feature map is then added to the previously obtained spatial–temporal correlated features to obtain the final interacted low-frequency component. As shown in the formula

$$\text{banch}_1 = \text{Conv}_{1\times1}(\text{ReLu}(LL_1 - LL_2)) \tag{7}$$
$$\text{banch}_2 = \text{Conv}_{1\times1}(\text{ReLu}(LL_2 - LL_1)) \tag{8}$$
$$\text{banch}_3 = \text{Conv}_{2\times3\times3}([LL_1, LL_2]) \tag{9}$$
$$LL = \text{banch}_3 + \text{Conv}_{3\times3}(\text{concat}[\text{banch}_1, \text{banch}_2]) \tag{10}$$

where concat[., .] represents the concatenation of features along the channel dimension.

After the interaction among the various frequency components of the bi-temporal features, the change feature maps are reconstructed from the interacted four frequency components through IDWT. The complete process of WIM is as described in the formula

$$f^i_{\text{order1}} = \text{WIM}\left(f^i_1, f^i_2\right)$$
$$f^i_{\text{order2}} = \text{WIM}\left(f^i_2, f^i_1\right) \tag{11}$$

where i represents the feature layer for the ith scale.

D. Feature Aggregation and Upsampling Module

The deep-level features contain abundant semantic information, while the shallow-level features contain a wealth of texture information [54], [55], [56]. Therefore, to further aggregate complementary information between feature maps at different scales and continuously enhance the representation capability of features, a FAUM is designed.

As shown in the bottom right corner of Fig. 2, FAUM consists of three branches, each corresponding to feature maps at three different scales: large, medium, and small. For the large-scale feature map F_l, representing the shallow-level features, boundary information is extracted using a max-pooling layer in the feature extractor. Subsequently, a 3×3 convolutional layer is employed to integrate this information while adjusting the channel dimension to match that of the medium-scale feature map. Regarding the medium-scale feature map F_m, information integration is performed using a 3×3 convolutional layer. As for the small-scale feature map F_s, its channel dimension is first adjusted to match the medium-scale feature map through a 3×3 convolutional layer, capturing semantic information. Subsequently, bilinear interpolation is utilized to upsample the features to the size of the medium-scale feature map. Next, the features of the three different scales are concatenated and aggregated to obtain a representative feature. As is shown in the formula

$$F'_l = \text{Conv}_{3\times3}(\text{MaxPool}(F_l))$$
$$F'_m = \text{Conv}_{3\times3}(F_m)$$
$$F'_s = \text{Bilinear}(\text{Conv}_{3\times3}(F_s)) \tag{12}$$
$$F = \text{Conv}_{3\times3}\left(\text{concat}\left[F'_l, F'_m, F'_s\right]\right). \tag{13}$$

In addition, to prevent excessive distortion of the extracted feature F, the medium-scale feature map is processed through a 1×1 convolutional layer. Subsequently, a residual connection is applied by adding this processed feature to F, resulting in the final output of FAUM. The process can be formulated as

$$F = \text{Conv}_{1\times1}\left(F'_m\right) + F. \tag{14}$$

It is worth noting that for the final FAUM, since there is no large-scale feature map available, only two branches are utilized, namely the small-scale feature map and the medium-scale feature map.

IV. Experiments

This section begins with an introduction to the datasets utilized in the experiments, followed by a detailed description of the experimental setup. Then, the results of different methods on each dataset are presented. Finally, ablation experiments and analyses are performed on various modules of FDINet.

A. Datasets

This article presents the evaluation of FDINet on three distinct datasets, namely WuHan University (WHU) [57],

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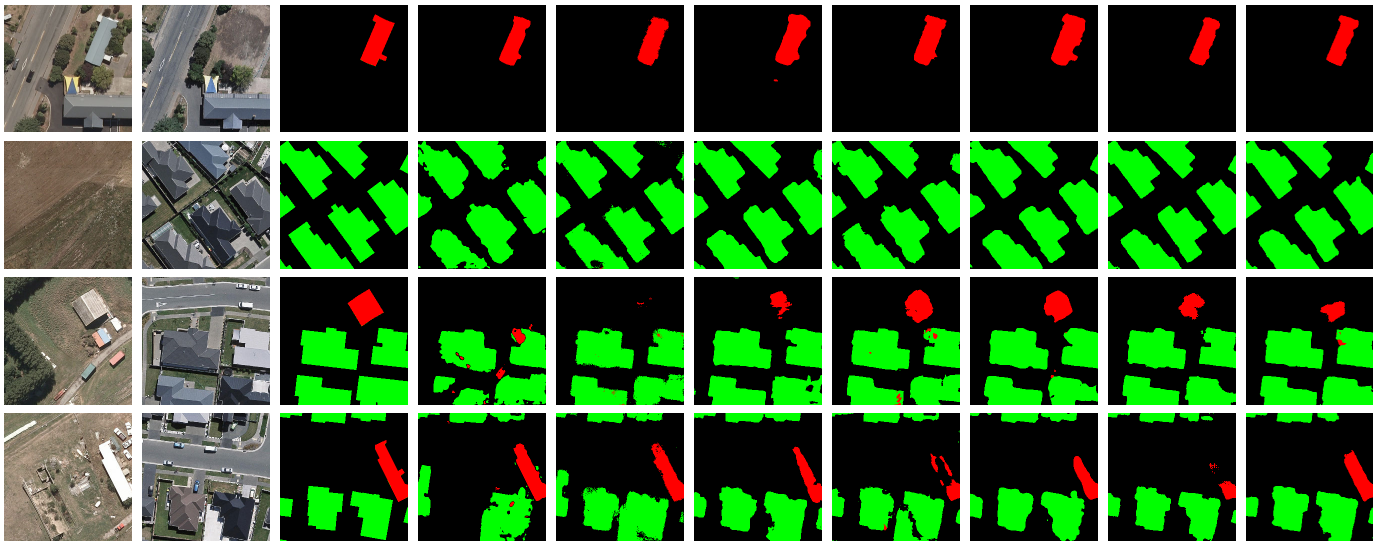


Fig. 6. Visual results of WHU. (a) and (b) Two temporal images; (c) ground truth; (d)–(i) prediction result of FC-EF, SNUNet, TINYCD, LSNet-dense, AFCF3D-Net, and FMCD; and (j) prediction result of FDINet. In the resulting images, green pixels represent the "appear" class and red pixels represent the "disappear" class.

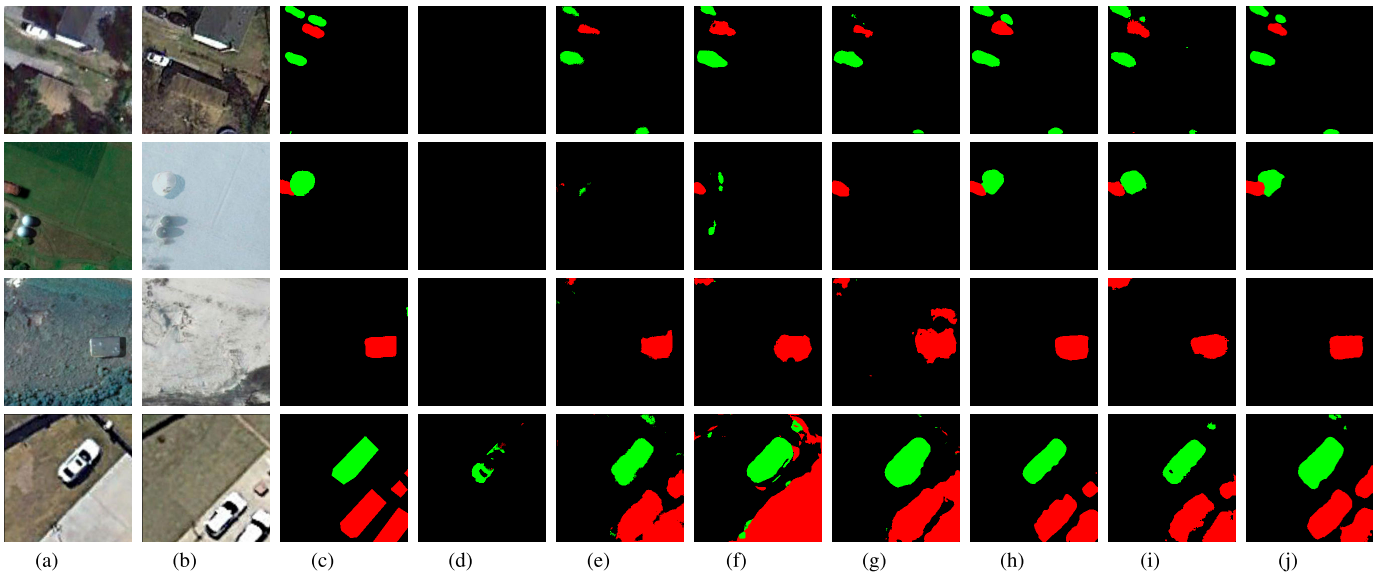


Fig. 7. Visual results of CDD. (a) and (b) Two temporal images; (c) ground truth; (d)–(i) prediction result of FC-EF, SNUNet, TINYCD, LSNet-dense, AFCF3D-Net, and FMCD; and (j) prediction result of FDINet. In the resulting images, green pixels represent the "appear" class and red pixels represent the "disappear" class.

change detection dataset (CDD) [58], and SECOND dataset [59].

WHU is a dataset focused on building change detection, with its original images sized at 32 507 × 15 354. The spatial resolution of the dataset is 0.3 m. To align with the requirements of OFCD tasks, we utilized the building labels of two temporal images provided by the original authors and reprocessed them to create OFCD data. The data was further cut into smaller patches of size 256 × 256. Ultimately, 5950 pairs of images were allocated for the training set, 742 pairs for the validation set, and 742 pairs for the test set.

CDD consists of real remote sensing images obtained via Google Earth, capturing seasonal variations. The resolution of the images ranges from 3 to 100 cm. The original data provided by the authors already includes OFCD labels. In this article, data augmentation was performed by applying horizontal flipping, vertical flipping, and rotations of 90°, 180°, and 270° to the images. Subsequently, the dataset was divided into 6444 pairs of images for training, 810 pairs for validation, and 810 pairs for testing.

SECOND is a semantic CCD that contains semantic information about all changing objects. To align with the OFCD task, we specifically extracted the semantic information related to buildings from the SECOND dataset. Accordingly, we established two change categories: "appear" for newly constructed buildings and "disappear" for demolished buildings. The original dataset consists of a total of 2968 pairs of images,

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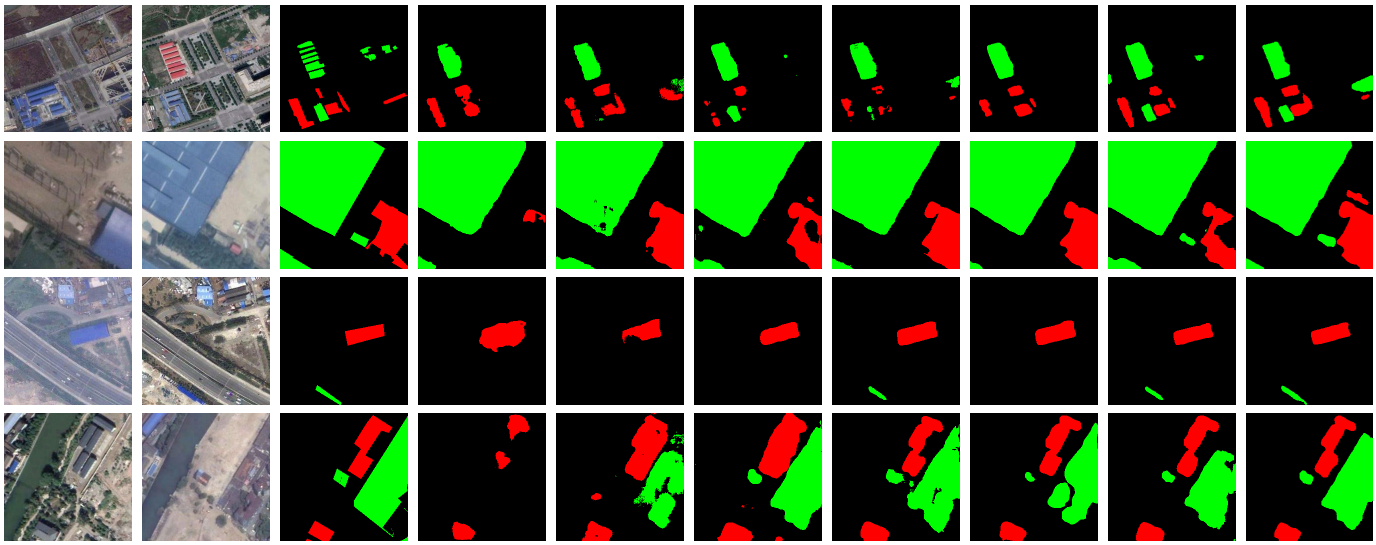


Fig. 8. Visual results of SECOND. (a) and (b) Two temporal images; (c) ground truth; (d)–(i) prediction result of FC-EF, SNUNet, TINYCD, LSNet-dense, AFCF3D-Net, and FMCD; and (j) prediction result of FDINet. In the resulting images, green pixels represent the "appear" class and red pixels represent the "disappear" class.

each with a size of 512 × 512. We cropped these images into 11 872 pairs of 256 × 256 image pairs. Subsequently, the data were divided into training, validation, and testing sets, with 9500, 1186, and 1186 image pairs, respectively.

B. Implementation Details and Evaluation Metrics

We conducted a series of experiments using PyTorch as our deep learning framework. The AdamW optimizer was employed for network optimization. The initial learning rate was set to 0.001, and it was dynamically adjusted using the cosine annealing scheduler. Due to hardware resource and memory constraints, the batch size was set to 32. During the training process, random flipping, rotation, and the addition of Gaussian noise were applied as data augmentation techniques. All training was performed on an NVIDIA GeForce RTX 3090 GPU.

Furthermore, we utilized the intersection over union (IoU) for each class as an evaluation metric to assess the accuracy of individual categories. The average IoU (mIoU) and average F1 (mF1) were employed as comprehensive evaluation metrics. The formulas for the evaluation metrics are provided as follows:

$$\text{Pre} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$
$$\text{Rec} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$
$$F1 = \frac{2 * \text{Pre} * \text{Rec}}{\text{Pre} + \text{Rec}} \tag{15}$$
$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{16}$$

where TP, FP, TN, and FN denote the quantities of true positives, false positives, true negatives, and false negatives, respectively.

C. Experimental Results

The differential-based methods tend to lose temporal correlation information, which is crucial for OFCD tasks. Hence, in this article, differential-based methods were not selected as a comparison method. In the experiments, a total of six state-of-the-art comparative methods, denoted as FC-EF [10], SNUnet [18], TINYCD [16], LSNet-dense [34], AFCF3D-Net [37], and FMCD [27], were adopted. Daudt et al. [10] and Liu et al. [34] both proposed different versions of a network, and the concatenated version was chosen among them to suit the OFCD task. Additionally, SNUNet captured temporal information by concatenating features along the channel dimension, while TINYCD and FMCD achieved temporal information extraction through feature fusion. On the other hand, AFCF3D-Net approached the bi-temporal image processing from a video perspective. To align with the OFCD task, the activation function of the model's final layer was modified to Softmax(). Additionally, the binary CE (BCE) loss function was replaced with the CE loss function. All other settings were kept as the default for the model.

The following pertains to quantitative and qualitative analyses of the experiment.

1) Qualitative Evaluation: The visual results of different methods on the WHU, CDD, and SECOND datasets are presented in Figs. 6–8. From the figures, it can be observed that FDINet exhibits superior overall performance, effectively detecting change regions and accurately discerning change types. In contrast, FC-EF struggles to detect changes when there is a significant stylistic disparity between the two temporal images, as evident in the second row and the third row of Fig. 7. Moreover, FC-EF, SNUNet, TINYCD, and LSNet-dense demonstrate relatively ambiguous detection boundaries, possibly due to the lack of integration between semantic information from deep features and texture information from shallow features. In contrast, AFCF3D-Net, FMCD, and FDINet incorporate a feature aggregation module

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within their networks, facilitating the fusion of complementary information from different scales and thus yielding more precise detection boundaries. It can be observed that apart from FMCD and FDINet, the remaining methods encounter difficulties in detecting smaller change regions effectively, as evident in the bottom-right corner of the second row image in Fig. 8. The reason behind this might be that FMCD narrows the gap between background parts in the two temporal images, thereby increasing the discrepancy between change and unchanged areas. Conversely, FDINet employs frequency decoupling interaction, facilitating differential interaction with high-frequency information such as textures. This interaction yields a constrained effect on semantic information at the same position. For regions with more complex scenes, such as the fourth row in Fig. 6, the first row in Fig. 7, and the fourth row in Fig. 8, FDINet effectively addresses them by employing frequency decoupling interaction to handle different frequency components of the images. Specifically, high-frequency information is utilized to determine the detection boundaries, low-frequency information is employed to decide the detection content, and mid-frequency information is utilized to further refine the detection results. Through such a collaborative division of labor, the model's detection capability is significantly enhanced.

2) Quantitative Evaluation: The quantitative analysis of various change detection methods on the three datasets is presented in Tables I–III. In these tables, the best results are denoted in red font, and the second-best results are denoted in blue font. It can be observed that FDINet exhibits superior overall performance on all three datasets. Compared to other algorithms, FDINet achieves mIoU improvements of 0.41%, 0.24%, and 1.07% on the WHU, CDD, and SECOND datasets, respectively. On the WHU and CDD datasets, it is evident that the IoU for the "disappear" category is lower compared to the "appear" category. This discrepancy can be attributed to the scarcity of instances belonging to the "disappear" class in these datasets, leading to the model having limited exposure to relevant information related to the "disappear" category during the training process. For the SECOND dataset, there is not a significant difference in the number of pixels between the two categories. Consequently, the disparity in IoU values between these categories is relatively small. The red font and blue font are mainly associated with AFCF3D-Net, FMCD, and FDINet. The reason behind this could be that these three methods effectively extract abundant temporal contextual information, which proves to be crucial for precise-level change detection. However, compared to the other two methods, FDINet achieves the best results by effectively utilizing frequency decoupling interaction, which allows separate interactions with different frequency components. This approach facilitates the comprehensive utilization of bi-temporal features and enables the extraction of their spatial–temporal contextual information.

3) Robustness Analysis: The robustness of FDINet to temporal perturbations is discussed and analyzed in this section. Specifically, the manipulation of the input sequence of images $\mathcal{T}_1$ and $\mathcal{T}_2$ is employed to alter the temporal order of the images. We conducted tests on all methods by reversing the order (exchanging the order of input bi-temporal images), and the results are presented in Table IV. Here, mIoU serves as a comprehensive metric for evaluating the strengths and weaknesses of the models. Therefore, it is used as the evaluation criterion in Table IV. From the table, it is evident that the detection accuracy of other methods has significantly decreased, and even OFCD tasks are challenging to achieve. However, under the temporal exchange framework, FDINet exhibits robustness in the temporal domain. Even with the changes in the input image's temporal order, it continues to deliver favorable results. It is worth noting that, compared to the other two datasets, the precision drop of each model on the SECOND dataset is relatively minor. This phenomenon could be attributed to the fact that the SECOND dataset contains two change categories with approximately similar pixel counts and exhibits less stylistic disparity between the bi-temporal images.

TABLE I
COMPARISON RESULTS ON WHU DATASET

Method	mF1	mIoU	IoU_{bg}	IoU_{da}	IoU_{ap}
FC-EF	0.7243	0.6369	0.9827	0.2079	0.7201
SNUNet	0.8460	0.7548	0.9864	0.5129	0.7652
TINYCD	0.8273	0.7441	0.9904	0.4024	0.8396
LSNet-dense	0.9066	0.8394	0.9927	0.6637	0.8617
AFCF3D-Net	0.9133	0.8497	0.9934	0.6808	0.8749
FMCD	0.9161	0.8558	0.9946	0.6690	0.9039
FDINet	0.9197	0.8599	0.9941	0.6943	0.8914

TABLE II
COMPARISON RESULTS ON CDD DATASET

Method	mF1	mIoU	IoU_{bg}	IoU_{da}	IoU_{ap}
FC-EF	0.6453	0.5384	0.9367	0.1768	0.5019
SNUNet	0.8128	0.7038	0.9577	0.5197	0.6339
TINYCD	0.7870	0.6723	0.9515	0.4689	0.5966
LSNet-dense	0.8123	0.7048	0.9595	0.4974	0.6575
AFCF3D-Net	0.8372	0.7359	0.9677	0.5653	0.6476
FMCD	0.8450	0.7450	0.9640	0.6044	0.6668
FDINet	0.8443	0.7474	0.9691	0.5479	0.7253

TABLE III
COMPARISON RESULTS ON SECOND DATASET

Method	mF1	mIoU	IoU_{bg}	IoU_{da}	IoU_{ap}
FC-EF	0.6529	0.5324	0.9194	0.2541	0.4238
SNUNet	0.8047	0.6927	0.9468	0.5052	0.6263
TINYCD	0.8206	0.7133	0.9515	0.5283	0.6602
LSNet-dense	0.8145	0.7050	0.9494	0.5254	0.6403
AFCF3D-Net	0.8359	0.7328	0.9559	0.5687	0.6738
FMCD	0.8192	0.7124	0.9532	0.5166	0.6673
FDINet	0.8437	0.7435	0.9579	0.6874	0.5772

D. Ablation Studies

In this subsection, the role of the temporal exchange framework is discussed, and the effectiveness of the WIM and the FAUM are analyzed. Throughout the entire ablation studies, mIoU is employed as the evaluation metric to assess the performance of each part of FDINet.

1) Effectiveness of the Temporal Exchange Framework: In order to validate the impact of the proposed temporal alignment framework, relevant ablation experiments were con-

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TABLE IV
EXPERIMENTAL RESULTS UNDER THE REVERSED TEMPORAL ORDER

	FC-EF	SNUNet	TINYCD	LSNet-dense	AFCF3D-Net	FMCD	FDINet
WHU	0.2116	0.3966	0.5857	0.2467	0.4733	0.5000	0.8532
CDD	0.2884	0.3858	0.3908	0.4085	0.4022	0.4095	0.7103
SECOND	0.4888	0.6596	0.6790	0.6871	0.7042	0.6581	0.7406

TABLE V
ABLATION EXPERIMENTS OF TEMPORAL EXCHANGE FRAMEWORK

Dataset		WHU	CDD	SECOND
original	FDINet	0.8599	0.7474	0.7435
	w/o	0.8555	0.7404	0.7316
revised	FDINet	0.8532	0.7103	0.7406
	w/o	0.6631	0.4294	0.7033

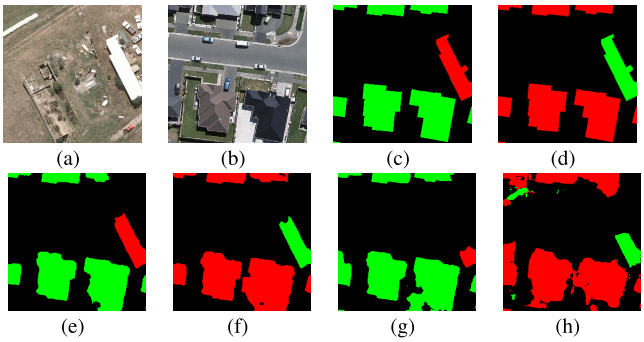


Fig. 9. Visualization results of ablation experiments of temporal exchange framework on WH. (a)–(d) T1 image, T2 image, ground truth map, and the ground truth map with the revised temporal order, (e) result of FDINet, (f) result of FDINet in the revised temporal order, (g) result of FDINet without temporal exchange framework, and (h) result of FDINet without temporal exchange framework in the revised temporal order.

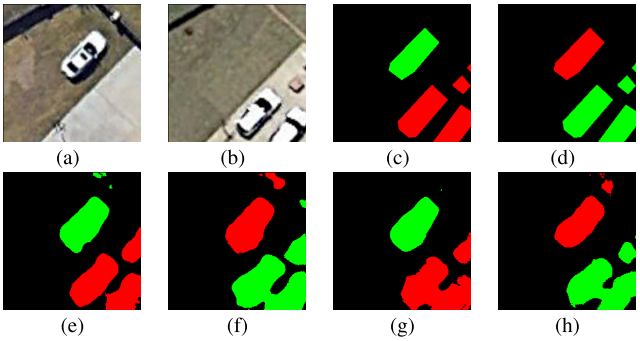


Fig. 10. Visualization results of ablation experiments of temporal exchange framework on CDD. (a)–(d) T1 image, T2 image, ground truth map, and the ground truth map with the revised temporal order, (e) result of FDINet, (f) result of FDINet in the revised temporal order, (g) result of FDINet without temporal exchange framework, and (h) result of FDINet without temporal exchange framework in the revised temporal order.

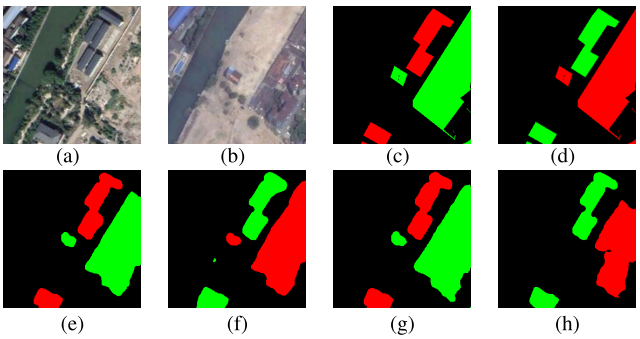


Fig. 11. Visualization results of ablation experiments of temporal exchange framework on SECOND. (a)–(d) T1 image, T2 image, ground truth map, and the ground truth map with the revised temporal order, (e) result of FDINet, (f) result of FDINet in the revised temporal order, (g) result of FDINet without temporal exchange framework, and (h) result of FDINet without temporal exchange framework in the revised temporal order.

ducted. Due to the framework's ability to improve robustness in the temporal domain, the experiments were conducted using two different temporal orders. The visual results of the ablation experiments for the temporal exchange framework are shown in Figs. 9–11. The experimental results are presented in Table V. In the table, "w/o" stands for "without," indicating the absence of the temporal exchange framework. "original" represents the test results of the model on the bi-temporal images in the normal temporal order. "revised" represents the test results of the model on the bi-temporal images in the reverse temporal order.

From the figures, it can be observed that the temporal exchange framework can enhance the model's performance, especially when it comes to improving the model's performance under the reversed temporal order. This indicates that the framework not only enhances the model's sensitivity to temporal variations but also significantly improves its robustness. From the table, it can be observed that the temporal exchange framework can enhance the model's capability for OFCD to a certain extent. Furthermore, even after exchanging the temporal order of input image pairs, FDINet still produces favorable results, indicating that the temporal exchange framework significantly improves the model's temporal domain robustness and enhances its ability to detect change types, aligning well with the task requirements. In addition, by comparing Tables IV and V, it can be observed that in the absence of the temporal exchange framework, FDINet continues to exhibit superior performance in the OFCD task. This indirectly indicates that FDINet effectively extracts temporal relevant information through its wavelet-based decoupled interaction mechanism.

2) Effectiveness of WIM: To validate the efficacy of the proposed WIM, three different interaction approaches were employed for comparison. The difference and concatenation approaches are classical feature interaction techniques. However, The difference approach in the method leads to the loss of temporal correlation in the interacted features. Hence, only concatenation was selected for comparison. Moreover, for OFCD tasks, temporal correlation information is crucial. Therefore, two novel 3-D convolution-based interaction

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TABLE VI
ABLATION EXPERIMENTS OF INTERACTION APPROACH

Interaction Approach	WHU	CDD	SECOND
Concatenation	0.8448	0.7301	0.7410
$Conv_{2\times1\times1}$	0.8482	0.7257	0.7407
$Conv_{2\times3\times3}$	0.8341	0.7379	0.7403
WIM	0.8599	0.7474	0.7435

TABLE VII
ABLATION EXPERIMENTS OF FAUM

Interaction Approach	WHU	CDD	SECOND
w/o	0.8548	0.7372	0.7345
FAUM	0.8599	0.7474	0.7435

approaches are designed for comparison. The first approach employs a $2 \times 1 \times 1$ convolutional kernel, similar to the mid-frequency interaction in WIM, to extract temporal information. The second approach follows 3-D convolutional branch of the low-frequency interaction in WIM, using a $2 \times 3 \times 3$ convolutional kernel to capture spatial–temporal correlation.

The results of the ablation experiments are presented in Table VI. Among the three compared interaction approaches, on the WHU dataset, the interaction approach using a $2 \times 1 \times 1$ 3-D convolutional kernel showed superior performance. This could be attributed to the higher resolution of the WHU dataset and the distinct changes between the two temporal images, resulting in clear boundaries of the changes. Therefore, focusing solely on extracting temporal correlation information yielded favorable results. On the other hand, for the CDD dataset, the interaction approach using a $2 \times 3 \times 3$ 3-D convolutional kernel achieved better results. This is because the CDD dataset exhibits significant differences between the two temporal data and has a lower resolution. As a result, it requires spatial structure assistance to achieve improved performance. Regarding the SECOND dataset, which lies between WHU and CDD in terms of data characteristics, the differences in performance among the three compared methods were not significant. Furthermore, FDINet utilizes frequency-domain decoupling to separate features into different frequency components, and each frequency component adopts a different interaction approach based on its characteristics. Consequently, FDINet with WIM achieved the best results among all interaction approaches.

3) Effectiveness of FAUM: In order to validate the roles of the FAUMs, ablation experiments were conducted. In the absence of the FAUM, the input from the WIM was directly concatenated with the data during the upsampling process. The results of the ablation experiments are presented in Table VII.

Based on the results in the table, it is evident that the model's performance improves when utilizing the FAUM. This enhancement can be attributed to FAUM's capability to aggregate texture information from large-scale feature maps and semantic information from small-scale feature maps into intermediate-scale feature maps. Consequently, it extracts beneficial features that are sensitive to detecting changes.

V. CONCLUSION

In this article, a frequency-decoupled interaction-based change detection method (FDINet) is proposed for OFCD tasks. In FDINet, a temporal exchange framework is proposed to improve the model's ability to differentiate between change types and enhance temporal robustness. Additionally, the WIMs are introduced to decouple features into multiple components based on their frequency, employing different interaction mechanisms for these components according to their frequencies. This facilitates the extraction of spatial–temporal correlation information vital for OFCD tasks. Subsequently, FAUMs are proposed to aggregate complementary information from different scale feature maps and progressively refine the detection results. Experimental results on three datasets show that FDINet outperforms the majority of state-of-the-art change detection methods. Furthermore, OFCD can accurately identify the change direction of the objects of interest in remote sensing images, which plays a crucial role in providing essential technical guidance for decision-making.

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